

Topic 14 Pearson

Ejercicios Propuestos

$$1) W = p(V_2 - V_1) = 500 \cdot 10^3 (0,71 \cdot 10^{-3} - 0,4 \cdot 10^{-3}) = 155 \text{ J}$$

$$2) \Delta U = 0 \quad Q = W$$

$$W = \int_{V_1}^{V_2} p(V) dV = p(V_2 - V_1) = 3646,8 \text{ J}$$

$$3) V_0 = 0,1^3 = 0,001 \text{ m}^3$$

$$V = V_0(1 + \beta \Delta T) \quad \beta = 3\alpha$$

$$V = 0,001 (1 + 1,7 \cdot 10^{-5} \cdot 3 \cdot (300 - 16)) = 1,014484 V_0 \\ = 1,01 \cdot 10^{-3} \text{ m}^3$$

$$W = p(V_2 - V_1) = 1,47 \text{ J}$$

$$4) P_{\text{tile}} = 4500 - 600 = 3900 \text{ W}$$

$$Q = P \cdot t = 3900 \cdot 3 \cdot 60 \cdot 60 = 4,21 \cdot 10^7 \text{ J}$$

$$5) Q = mL_v = 0,005 \cdot 2,26 \cdot 10^6 = 11300 \text{ J}$$

$$n = \rho V \rightarrow V = \frac{m}{\rho}$$

$$V_2 = \frac{5}{5,98 \cdot 10^{-4}} = 8361,2 \text{ cm}^3 = 8,36 \cdot 10^{-3} \text{ m}^3$$

$$V_1 = \frac{5}{0,958} = 5,219 \text{ cm}^3 = 5,219 \cdot 10^{-6} \text{ m}^3$$

$$W = p(V_2 - V_1) = 1,01 \cdot 10^5 (8,36 \cdot 10^{-3} - 5,219 \cdot 10^{-6}) = 843,95$$

$$\Delta U = Q - W \quad \Delta U = 11300 - 843,95 = 10456,5$$

$$6) n = 0,5 \quad V_1 = 15 \cdot 10^{-3} \text{ m}^3 \quad T_1 = 352 \text{ K}$$

$$V_2 = 7,9 \cdot 10^{-3} \quad T_2 = 291 \text{ K}$$

Camino 1:

1 → 2: Isocórico → $W = 0$

$$P_1 V_1 = n R T_1 \rightarrow P_1 = \frac{0,5 \cdot 8,314 \cdot 291}{7,9 \cdot 10^{-3}} = 1,53 \cdot 10^5 \text{ Pa}$$

2 → 3: Isobárico →

$$W = P_3 (V_3 - V_2) = 1,53 \cdot 10^5 (7,9 \cdot 10^{-3} - 15 \cdot 10^{-3})$$

$$W = -1087,25$$

$$W_T = W_3$$

Camino 2:

1 → 2: Isobárico

$$\frac{V_1}{T_1} = \frac{V_2}{T_2} \rightarrow \frac{15 \cdot 10^{-3}}{352} = \frac{V_2}{291} \rightarrow V_2 = 12,4 \text{ dm}^3$$

$$P_1 = \frac{0,5 \cdot 8,314 \cdot 352}{15 \cdot 10^{-3}} = 9,76 \cdot 10^4 \text{ Pa}$$

$$W = 9,76 \cdot 10^4 (12,4 - 15,4) \cdot 10^{-3} = -253,5775$$

2 → 3: Isotérmico

$$W = n R T \ln\left(\frac{V_2}{V_1}\right) = 0,5 \cdot 8,314 \cdot 291 \cdot \ln\left(\frac{7,9}{12,4}\right) = -545,43$$

$$W_T = -253,577 - 545,43 = -799 \text{ J}$$

